

LESSON 2-3 • SESSION 1 • GUIDED NOTES

Median and Outliers

Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.

• 6.SP.B.5d

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
outlier <i>valor atípico</i>	A value ____ from all the others.	120 in 39, 41, 43, 44, 45, 47, 120
median <i>mediana</i>	The ____ value of the ordered data.	44
mean <i>media</i>	The sum of the values ____ by how many values there are.	$(2 + 4 + 9) \div 3 = 5$

KEY IDEA*complete each sentence*

- An outlier pulls the ____ toward it.
- The median stays in the ____, so it describes typical data better when there is an outlier.

WORKED EXAMPLE*fill in the blanks as we solve*

Car counts: 44, 39, 41, 47, 43, 45, 120. What is a typical count?

Step 1 Order: 39, 41, 43, 44, 45, 47, 120. The outlier is ____.**Step 2** The median is the 4th value: ____.**Answer** A typical Saturday has about ____ cars.**YOU TRY***use the example as a model***1.** Data: 12, 15, 11, 14, 13, 60, 16.**a.** Outlier: _____**b.** Median: _____**2.** In the data above, 60 changes to 90. What is the new median?**Median** _____**SUM IT UP***in your own words*

When a data set has an outlier, I use the ____ because ____.

LESSON 2-3 • SESSION 2 • GUIDED NOTES

Comparing Distributions

Learning target – I can compare two distributions using the median and the interquartile range together. · 6.SP.B.5d

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
distribution <i>distribución</i>	How the data values are _____ across their range.	a box plot shows it
center (median) <i>centro (mediana)</i>	The _____ value of a data set.	median wait 8 min
spread (IQR) <i>dispersión (rango intercuartil)</i>	How _____ the middle half is.	IQR 7 min

KEY IDEA*complete each sentence*

- Use the _____ to compare typical values and the IQR to compare consistency.
- A higher median does not mean _____; only the IQR shows that.

WORKED EXAMPLE*fill in the blanks as we solve*

Line 1: median 8 min, IQR 7. Line 2: median 13 min, IQR 12. Which line should you join?

Step 1 Center: Line 1's typical wait is ____ min shorter.**Step 2** Spread: Line 1's IQR is _____, so its waits are steadier.**Answer** Join Line ____.**YOU TRY***use the example as a model***1.** Group 1: median 80, IQR 4. Group 2: median 84, IQR 22.

- a.** Higher typical score: _____ **b.** More consistent: _____

2. Team A: median 30, IQR 8. Team B: median 44, IQR 8. How are they alike? How do they differ?**Alike; differ** _____**SUM IT UP***in your own words*

To compare two distributions, I use the median to ____ and the IQR to ____.
